Submission Quality Score

§1 — Structure: the six pillars and their weights

Each pillar produces its own 0–100 sub-score. The submission score is their weighted sum.

Pillar	Weight	What it measures
Structural Completeness	25%	Core application data: producer, applicant, address, dates, lines of business, entity type, and how completely the ACORD forms filled
Exposure Consistency	25%	Class codes, payroll and revenue alignment, coverage matched to actual exposure
Property Integrity	15%	COPE completeness: Construction, Occupancy, Protection, Exposure
Loss History Alignment	15%	Loss runs, claim history, and whether they reconcile with the stated exposures
Umbrella & Limit Adequacy	10%	Underlying limits against the umbrella attachment point
Narrative Quality	10%	Underwriting context and account story

> *Umbrella is the one nullable pillar. If the submission carries no umbrella, that pillar is marked Not Applicable, removed from the calculation entirely, and the remaining five weights are scaled up proportionally. It is never scored as zero.*

§2 — Calculation: raw score, then ceiling, then credits

Standard case

```
Raw SQS = (Structural × 0.25)
        + (Exposure   × 0.25)
        + (Property   × 0.15)
        + (Loss       × 0.15)
        + (Umbrella   × 0.10)
        + (Narrative  × 0.10)
```

Umbrella not applicable

```
Scale   = 1 ÷ (1 - 0.10) = 1.1111

Raw SQS = [ (Structural × 0.25)
          + (Exposure   × 0.25)
          + (Property   × 0.15)
          + (Loss       × 0.15)
          + (Narrative  × 0.10) ] × 1.1111
```

Effective weights become:

Structural	27.8%	Exposure	27.8%	Property	16.7%
Loss	16.7%	Narrative	11.1%		

Worked example — GL and property, no umbrella

Structural 88 · Exposure 92 · Property 75 · Loss 60 · Narrative 50

```
(88 × 0.278) + (92 × 0.278) + (75 × 0.167)
  + (60 × 0.167) + (50 × 0.111)
= 24.5 + 25.6 + 12.5 + 10.0 + 5.6
= 78 raw
```

The result is rounded down to a whole number and bounded to 0–100. Then, in this order:

```
Ceiling       = 60 if any hard stop
                85 if no hard stop but any warning
                none otherwise

Displayed SQS = minimum of ( Raw SQS + earned credits , Ceiling )
```

The raw score is kept, not overwritten. Every submission stores its uncapped score, the ceiling applied, and the specific condition that caused it. A capped score always reads as one sentence: “88 raw, held at 60 by an unresolved FEIN conflict.”

This is also why later improvements compound off the real number. A submission at raw 65 with a hard stop displays 60 but stores 65. Clear the stop and earn 10 more points and it reads 75, not 70.

§3 — Drivers: what moves each pillar

3.1 Structural Completeness — 25%

Submission level:

```
Structural   = (Tier 1 score × 0.35)
             + (Tier 2 score × 0.30)
             + (Fill rate    × 0.35)

Tier 1 score = 100 if all present, else 100 - (20 × number missing)
Tier 2 score = 100 - (number missing × 100 ÷ number applicable)
Fill rate    = confidence-weighted percentage of ACORD fields that
               received a value, averaged across all generated forms
```

Tier 1 — six fields plus contact

Producer / agency name	Applicant legal name
Applicant mailing address	Proposed effective date
Lines of business requested	Business entity type
Any one of contact name, phone or email	

Producer name is not required when the only source document is a declarations page. Certificate-only submissions are judged on two items instead: applicant legal name and effective date.

Tier 2 — twelve fields

FEIN / Tax ID	Operations description
Annual revenue	Prior carrier name
Number of employees	Years in business
NAICS or SIC code	Number of prior claims
Annual payroll	WC experience modification factor
WC payroll period	WC owner / officer exclusions

NAICS and SIC are interchangeable — either one satisfies the industry classification requirement. The three Workers Comp fields drop out of both the missing list and the denominator when the submission has no WC coverage, so a complete non-WC submission can reach 100. If no forms have been generated yet, the fill-rate component is dropped and the other two rescale to 53.8% and 46.2%, holding their original ratio.

Individual form scores use that form's own checklist

Form	Required items, equally weighted
ACORD 125	Applicant name, mailing address, effective date, lines of business, contact info, producer name
ACORD 126	GL limits, GL class codes, operations description, payroll or revenue, GL form type

Form	Required items, equally weighted
ACORD 140 / 141	Minimum Viable COPE. If incomplete the score is 0; otherwise 60 + 5 per Tier-1 COPE field + 5 per Tier-2 COPE field
ACORD 127	Vehicle schedule, auto liability limit, garaging address, driver schedule, covered auto symbols, radius of operation
ACORD 130	Payroll, WC class codes, X-mod, prior carrier, officer exclusions, employee count
ACORD 131	Umbrella limit, SIR or attachment point, GL limits, auto liability limit, Employers Liability limits
ACORD 28	Applicant, both policy dates, policy number, building or BPP value, mortgagee or certificate holder
ACORD 101	Applicant, effective date, form reference, remarks — plus 10 bonus points for substantive remarks
ACORD 25	Applicant or certificate holder, effective date, policy number, GL limits
133 / 137 / 138 / 160 / 186	Form-specific checklists on the same equal-weight model

> *OCR confidence penalty, individual forms only. Minus 6 points for each Tier-1 field the system could not read with confidence, capped at minus 30.*

3.2 Exposure Consistency — 25%

Starts at 100. Deductions accumulate into five visible buckets. The pillar score is 100 minus the total.

Bucket	Condition	Points
Operations	GL coverage with no class codes at all	-20
	Class codes present but no operations description	-10
	GL class code does not match the operations description	-15
	No GL coverage and no operations description	-10
	WC coverage with no class codes	-10
	WC class code does not match the operations description	-15
	Contractor or subcontractor exposure with no subcontracted percentage	-8
Coverage	GL coverage with no limits	-8
	Auto coverage with no liability limit	-10
	Auto coverage with no covered-auto symbols	-5
	Vehicles found in the documents but no auto coverage requested	-8
Payroll & employees	No payroll and no revenue anywhere in the submission	-15

Bucket	Condition	Points
	WC coverage with no payroll	-12
	Multi-state WC with no payroll breakdown by state	-8
	More than 5 employees and no WC coverage	-8
Revenue & sales	Payroll above 200% of revenue	-10
	Payroll below 1% of revenue	-5
Cross-document	Graduated by volume of cross-form conflicts	up to -20

Cross-document bucket:

```
Deduction = minimum of ( 15 × hard cross-form issues
                       +  5 × cross-form warnings , 20 )
```

3.3 Property Integrity — 15%

```
No property coverage in the submission  = 100  (Not Applicable, scored clean)
Minimum Viable COPE incomplete          = 0    (also a hard stop)
Otherwise = 60 + (5 × Tier-1 fields) + (5 × Tier-2 fields)
ACV vs RCV valuation conflict           = -10
```

Set	Fields	Each
Minimum Viable COPE	Location address · Occupancy type · Construction type · Building value or Business Personal Property value	required
Tier 1	Year built · Roof year · Sprinkler system · Fire protection class	+5
Tier 2	Distance to hydrant · Fire department type · Business income limit · Period of restoration	+5

Valuation method is deliberately excluded from the Tier-2 credits and handled as a separate validation rule instead. Counting it as a credit field permanently capped a fully carrier-grade submission at 96.

On non-property forms that still carry property exposure: minus 5 for no valuation method, minus 5 for no coinsurance percentage.

3.4 Loss History Alignment — 15%

Path A — loss runs uploaded with readable claim years

Years of loss runs	Base score
5 or more, fully valued	100
3 to 4	80
1 to 2	40

Then these adjustments apply:

Condition	Points
Prior carrier name present	+10
Prior carrier name missing	-10
Loss runs 0 to 90 days old — currently valued	0
Loss runs 91 to 180 days old	-10
Loss runs 181 to 365 days old	-15 rising to -20
Loss runs over 365 days old	-25
Valuation date cannot be determined at all	-15
Insured match: name + FEIN or policy number	0
Insured match: name + address	-8
Insured match: name only	-15
Insured match: no match	-30
Claim frequency above 2 claims per $1M of exposure	-25
Claim frequency above 1 claim per $1M of exposure	-10
Loss ratio above 10% of exposure	-15

Path B — loss runs uploaded but no readable claim years

Ownership match quality	Base score
Strong — name plus FEIN or policy number	60 fixed
Moderate — name plus address	42
Possible — name only	35
No match	15

Prior carrier ±10 and the recency rules still apply to moderate, possible and no-match. The strong tier is pinned.

Path C — no loss runs at all

Situation	Score
Loss runs requested or pending	70
No known losses, attested by the insured	60
No known losses, mentioned only in narrative prose	45
Nothing provided	25

> *Two hard ceilings on this pillar. Loss runs that do not match the insured cap it at 25 — unmatched runs are not creditable evidence for this submission. A no-loss attestation contradicted by actual claims in the uploaded loss runs caps it at 45.*

3.5 Umbrella & Limit Adequacy — 10%

```
No umbrella in the submission              = Not Applicable
                                             (pillar removed, weights rescaled)
Umbrella with no underlying GL and no Auto = 0 (also a hard stop)
Otherwise = 100 minus the deductions below
```

Condition	Points
The umbrella’s own limit is missing	-25
GL each-occurrence below $1,000,000, or required and absent	-20
GL aggregate below $2,000,000	-20
Auto combined single limit below $1,000,000, or required and absent	-20
Employers Liability missing when WC is present	-25
Employers Liability between $500,000 and $999,999	-10
Employers Liability below $500,000	-25
No Schedule of Underlying Insurance	-15
Follow-form status not explicitly and affirmatively confirmed	-10

> *Underlying-limit shortfalls reduce the score, they never hard stop. Carrier attachment points vary by market, so this is a warn-and-reduce rule by design.*

3.6 Narrative Quality — 10%

```
Narrative = ( Component coverage %    × 0.60
            + Underwriting substance % × 0.40 ) × Source factor

Component coverage % = components present ÷ 12 × 100
```

Twelve components, each present or absent

Account Overview	Operations Description
Years in Business	Management Experience
Risk Controls	Loss History Discussion
Coverage Discussion	Prior Carrier / Marketing Reason
Location Details	Employee & Payroll Context
WC Payroll & Class Code Context	EMOD / XMOD Information

Modifier	Condition	Effect

Source factor	A real uploaded narrative document, or structured remarks	× 1.00
Source factor	Only the operations description, with no real narrative	× 0.75
Floor	A genuine narrative document is present in the package	min 40
Bonus	Submission urgency or deadline supplied via questionnaire	+8
Baseline	No narrative content of any kind	0

Substance scores whether the narrative gives an underwriter usable context, not how long it is. Length and vocabulary range are explicitly not rewarded.

A short curated field credits a component on a single clear mention. A long scanned narrative must contain at least two distinct signals for a component, so stray words in letterhead or boilerplate cannot inflate the score.

§4 — Conditions: complete hard-stop list

Hard stops come from three independent checks. All three feed the same ceiling of 60.

Field-level — 7 conditions

1. Effective date falls on or after the expiration date — the policy period is invalid
2. Minimum Viable COPE incomplete: missing location, occupancy type, construction type, or building / BPP value
3. Peril-specific deductibles referenced but never defined (wind and hail, earthquake, flood)
4. Monopolistic WC state (ND, OH, WA, WY) with private-carrier WC requested and no state-fund acknowledgment
5. Monopolistic WC state detected but no monopolistic payroll breakdown provided
6. Umbrella coverage detected but no underlying GL or Auto limits found
7. Split auto liability limits incomplete — all three of BI per person, BI per accident and PD per accident are required

Cross-document identity conflicts — 4 conditions

1. Applicant legal name differs across uploaded documents
2. FEIN differs across uploaded documents
3. Effective date differs across uploaded documents
4. Expiration date differs across uploaded documents

> *Items 3 and 4 downgrade to warnings when a document carries an explanation that specifically addresses the policy period. Comparison is normalization-aware: 07/15/25 against 7/15/2025, or "LLC" against "L.L.C.", never triggers a conflict. Any value already confirmed in the Data Consistency panel is excluded.*

Cross-form conflicts — 11 conditions

1. WC payroll does not reconcile with total payroll
2. Location count differs between the documents and the forms
3. Business Income coverage present with no Period of Restoration
4. Multi-state WC with no payroll breakdown by state
5. State-by-state WC payroll does not sum to the stated total
6. Umbrella present with no underlying coverage
7. Owned fleet not covered by the liability symbol on file — the fleet has no liability coverage
8. Split auto liability limits incomplete
9. Heavy subcontracting exposure with no WC payroll
10. Peril deductible referenced but undefined
11. Minimum Viable COPE missing

Hard only when unexplained — 5 conditions

- Umbrella effective date does not match the GL policy effective date
- Umbrella expiration date does not match the GL policy expiration date
- Umbrella effective date does not match the auto policy effective date
- Umbrella expiration date does not match the auto policy expiration date

- Umbrella period does not match the Workers Comp period

These become warnings when an ACORD 101 narrative explains the difference.

Submission level only — 1 condition

- Building value flagged for review because it differs across documents and has not been confirmed

Individual form only — 3 conditions

- Property Integrity at zero on ACORD 140, 141 or 133
- Umbrella pillar at zero with an umbrella present
- Business Income limit present with no Period of Restoration, or peril deductibles referenced but undefined

These cap that form's own score. They never cap the submission score.

> *Every cross-form rule is gated on the relevant ACORD form actually being selected. A rule for a form the broker did not choose never fires.*

§5 — Conditions: complete warning list

Warnings hold the score at 85 when no hard stop is present.

Format and range

Mailing address missing a valid US state or ZIP · Phone number not 10 or 11 digits · Email address malformed · FEIN not exactly 9 digits · Effective or expiration date unparseable · Subcontracted percentage outside 0–100 · Coinsurance percentage outside 0–100 · Building ITV percentage outside 0–100 · Negative value on any of revenue, payroll, WC payroll, building value, BPP value, umbrella limit, GL each occurrence, GL aggregate, auto liability limit or business income limit

Underwriting

Effective date outside the acceptable window · Invalid NAICS code prefix · Policy term already expired or expiring · Prior carrier adverse action with no narrative explanation · Claims-made GL with no retroactive date · GL coverage with no revenue or payroll found · GL coverage with no class codes found · Carrier-grade COPE incomplete · Business Income coverage with no BI limit · Valuation method not specified · ACV and RCV both appear across the documents · Payroll above 200% of revenue · Payroll below 1% of revenue · WC coverage with payroll missing · Monopolistic WC state acknowledged · Multi-state WC with no state breakdown · Physical damage coverage with no deductibles specified · Underlying auto limits may not meet umbrella requirements · Umbrella over WC with no Employers Liability limits · Employers Liability below the $500,000 market minimum · Umbrella and GL policy periods misaligned · Umbrella and GL expiration dates misaligned

Cross-form — 37 conditions

WC payroll implausible against stated revenue · Subcontracting percentage conflicts with WC payroll · GL class codes with no operations description · Contractor risk with no Contractors Supplemental form · Minor location-count difference · Location addresses differ across forms · Umbrella over WC with no Employers Liability · Employers Liability below the umbrella minimum · Same structure appears on both Builders Risk and Property · Business Income indicated with no limit · Baseline commercial application not included in the package · Crime exposure present with no crime coverage requested · Cyber exposure present with no cyber coverage requested · Hired and non-owned auto exposure not designated by a covered-auto symbol · No covered-auto symbols captured at all · Physical damage requested with no comprehensive or collision symbol · Drive Other Car indicated with no driver record · Property valuation method not stated · High subcontracting alongside high WC payroll · WC and GL class codes describe different operations · Claims-made GL with no retroactive date · Claims-made GL with no prior-acts coverage stated · Underlying GL limit below the umbrella attachment baseline · Underlying GL limits not found · Underlying auto limit below the umbrella baseline · Underlying auto limits not found · All-other-perils deductible missing · Peril deductible schedule incomplete · Deductible basis not stated · Coinsurance percentage missing · Coinsurance percentage outside a reasonable range · Physical and mailing address cannot be distinguished · Carrier-grade COPE incomplete · COPE incomplete at one or more locations · Agreed-value auto coverage with no vehicle schedule · Certificate requested but no ACORD 25 in the package · Property evidence requested but no ACORD 28 in the package

Cross-document — 5 conditions

DBA name differs across documents · Entity type differs across documents · Mailing address differs across documents · Physical address differs across documents · Lines of business differ across documents

Advisories — no score effect, 8 conditions

Inland Marine and Property may cover the same item · ACV valuation on a high-value building · RCV valuation on an older building · Uninsured / Underinsured Motorist not specified · PIP or Medical Payments not specified · Drive Other Car not specified · ACORD 101 recommended for overflow narrative · Legal name and DBA are identical

There is a deliberate third severity. These render as cards for the broker's attention but do not cap the score and do not deduct points.

§6 — Ceilings: how the 60 and 85 ceilings are applied

```
Any hard stop present                → ceiling 60
No hard stop, any warning present    → ceiling 85
Neither                              → no ceiling
```

Situation	Displays	Note
Raw 88, one hard stop	60	
Raw 88, warnings only	85	
Raw 42, one hard stop	42	the ceiling is not a floor
Raw 93, clean	93	

Three defining properties

- It is a ceiling, never a floor. A submission that genuinely scores 42 with a hard stop stays at 42. It is never lifted to 60. Two submissions at 22 and 56 under the same hard stop stay at 22 and 56 — they do not collapse onto a shared number.
- Hard takes absolute precedence. One hard stop plus fourteen warnings gives a ceiling of 60, not 85.
- Only the displayed number is capped. The uncapped score, the ceiling applied, and the condition responsible are all recorded against the submission.

Neither hard stops nor warnings block anything. Nothing in the product prevents form generation or download because of them. Their entire effect is the ceiling plus the card on screen.

§7 — Ceilings: when several conditions exist at once

They do not stack. One hard stop and fifteen hard stops produce exactly the same ceiling of 60. There is no cumulative penalty, no per-stop deduction, no escalation. The submission records the first hard stop as the stated reason for the cap, purely so the number is explainable. Every stop still renders its own card on screen.

Why they do not stack: the pillars already carry the severity. Five missing COPE fields score lower on Property Integrity than one missing field, and that difference survives intact in the raw score. Stacking cap penalties on top would count the same gap twice.

The one place volume does matter is the cross-document bucket inside Exposure Consistency, where cross-form issues carry 15 points per hard issue and 5 per warning, capped at 20 points total. That is a pillar deduction, not a ceiling.

§8 — Behaviour: clearing a condition recalculates automatically

Recalculation is a full rebuild from the current data, not a patch applied to the old number.

Triggers

- The client submits questionnaire answers
- The producer answers a recommendation
- The producer resolves an issue inline by entering the missing value
- The producer reopens a dismissed recommendation
- The producer undoes an applied resolution
- A PDF is regenerated, or a field is edited in the form editor

What each recalculation runs, in order

1. All field-level validations
2. All cross-document identity checks
3. All 33 cross-form rules
4. All six pillars, for every generated form
5. All six pillars, for the submission
6. Ceiling resolution
7. Re-application of any outstanding earned credits

> *One correction worth knowing about. The cross-document identity checks originally ran only at upload time, so the first recalculation quietly dropped an unresolved FEIN or name conflict from the ceiling list while its card stayed visible. The score rose because a blocker stopped counting, not because anything was fixed. That is closed — those checks now re-run on every recalculation.*

§9 — Behaviour: why marking an item Resolved or Dismissed does not change the score

There are three distinct actions available, and only two of them are meant to move the number.

Action	Score effect	Why
Resolved / Dismissed toggle on an issue card	NONE	Work tracking. It records that a broker has reviewed the item and decided what to do about it.
Dismiss a recommendation with a written reason	EARNS CREDIT	The stated point value is credited. A plain dismissal with no reason is worth zero.
Resolve the issue by entering the value	FULL RECALCULATION	The rule stops firing, the ceiling lifts, and the pillar earns its points.

The toggle, in detail

The reason is structural. A hard stop is derived from the data every time the score is calculated, it is not a stored flag. On each recalculation the engine re-reads the submission and re-tests the condition. Marking a card “Resolved” records a status; it does not change the FEIN, supply the missing sprinkler data, or add the Period of Restoration. So the condition still evaluates true, the stop still fires, and the ceiling still binds.

If acknowledgment lifted the ceiling, the score would stop meaning “this submission is carrier-ready” and start meaning “someone clicked a button.” A carrier receiving a 92 would have no way to tell the two apart.

How credits behave

- They are added to the raw score and then re-capped, so clearing the stops later releases the full earned value
- They are stored as an absolute total, so recalculations re-apply them rather than erasing them
- They are scoped per form by which recommendations that form actually carries; the submission score uses the full total
- They retire automatically once the underlying field is genuinely filled, because the pillars then award those points directly — the credit must not stack on top of the improvement it was standing in for

§10 — Scope: individual form scores against the total submission score

They use the same calculation. Same six pillars, same weights, same Not Applicable rescaling, same 60/85 ceiling rule, same tier ladder, same raw-score preservation.

	Individual form score	**Total submission score**
Structural input	That form's own ACORD checklist, minus an OCR confidence penalty of up to 30 points	Tier 1 × 0.35 + Tier 2 × 0.30 + fill rate × 0.35
Which stops apply	Field-level stops only. Cross-form conflicts never cap an individual form	Field-level and cross-form stops
Extra ceilings	COPE at zero on 140/141/133, umbrella at zero, form-level property gate	None
Cross-form penalty	Not applied	Feeds the Exposure Consistency cross-document bucket

> *The submission score is calculated independently. It is not an average of the form scores. Every pillar is derived directly from the merged submission data by submission-level calculators.*

That independence is why the submission score legitimately differs from a single form's score even when only one form is in the package: a different Structural formula, none of the per-form structural gates, and cross-form factors no individual form can see.

Tier ladder — identical for forms and submissions

Score	**Grade**	**Tier**
90 – 100	A	Submission Ready
80 – 89	B	Almost There
70 – 79	C	Needs Work
60 – 69	D	Major Gaps
0 – 59	F	Not Ready

A score of exactly 90 is Submission Ready.

Routing recommendation on individual forms

Score	Routing
Above 85	Auto quote
70 – 85	Priority review
50 – 69	Standard review